

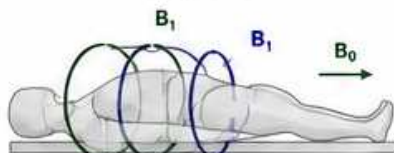
# MRI B<sub>1</sub> FIELD & FLIP-ANGLE CALIBRATION

Accurate B<sub>1</sub> mapping and flip-angle calibration ensure consistent tissue excitation, uniform images and quantitative reliability.

Uniform Excitation • Accurate Flip Angles • Consistent Image Contrast • Quantitative Accuracy • Better SNR

## 1. WHAT IS B<sub>1</sub>?

B<sub>1</sub> is the RF magnetic field (in the transverse plane) created by the transmit coil.  
It determines the flip angle ( $\alpha$ ) achieved in the tissue.



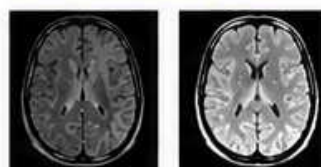
### Flip Angle ( $\alpha$ ) Relationship

$$\alpha = \gamma \int B_1(t) dt$$

- $\gamma$  = gyromagnetic ratio
- $\int B_1(t) dt$  = RF pulse area (in  $\mu T \cdot ms$ )

## 2. WHY B<sub>1</sub> VARIES

- Coil design & geometry
- Patient size, shape & position
- Dielectric effects & tissue properties
- Loading (detuning) from patient
- Frequency (off-resonance)
- System & amplifier performance



Under-flip ( $\alpha$  too low)      Over-flip ( $\alpha$  too high)

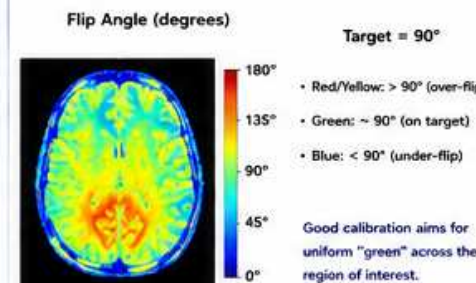
Result: Signal loss, non-uniform image intensity and quantitative errors.

## 3. B<sub>1</sub> MAPPING METHODS (COMMON)

| Method                          | Type   | Principle                                               | Pros                                         | Cons                                               |
|---------------------------------|--------|---------------------------------------------------------|----------------------------------------------|----------------------------------------------------|
| Actual Flip Angle (Ernst)       | Direct | Two images with different TRs<br>Solve for $\alpha$     | Quantitative $\alpha$ map<br>Relatively fast | TR sensitive<br>Requires T <sub>1</sub> assumption |
| Double Angle (DA)               | Direct | Images at $\alpha$ and $2\alpha$<br>Compute $\alpha$    | Simple<br>No T <sub>1</sub> needed           | Sensitive to noise<br>SAR higher                   |
| Bloch-Siegert (BS)              | Direct | Off-resonance pulse induces phase shift $\propto B_1^2$ | Independent of T <sub>1</sub><br>Low SAR     | Longer scan time<br>Lower SNR                      |
| AFI (Actual Flip-angle Imaging) | Hybrid | Uses transient state & Bloch simulation                 | Good accuracy<br>Robust                      | More complex<br>Longer time                        |
| pTx / DREAM                     | Direct | Multi-channel B <sub>1</sub> mapping                    | Accounts for channel interactions            | Advanced<br>Requires pTx                           |

Direct methods measure flip angle ( $\alpha$ ) directly.  
Hybrid methods estimate  $\alpha$  using modeling.

## 4. FLIP ANGLE MAP EXAMPLE (Actual Flip Angle)



- Typical Targets**
- 90° for spin echo
  - 180° for refocusing
  - Variable for specific sequences (GRE, etc.)

## 5. HOW CALIBRATION WORKS

- Perform B<sub>1</sub> mapping of the region of interest.
- Compare measured flip angle ( $\alpha_{meas}$ ) with target ( $\alpha_{target}$ ).
- Compute correction factor (CF):

$$CF = \frac{\alpha_{target}}{\alpha_{meas}}$$

- Adjust transmit gain (or pulse amplitude) by CF.
- Re-measure to verify uniformity.

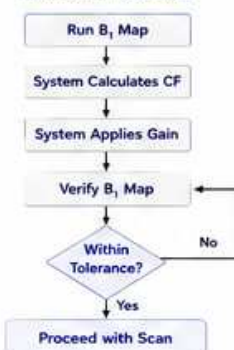
✓ Goal: Achieve  $\alpha_{meas} \approx \alpha_{target}$  ( $\pm 5-10\%$ ) across the imaging volume.

## 6. CALIBRATION WORKFLOWS

### Manual (Gain Adjust)



### Auto (System Assisted)



## 7. GOOD vs POOR CALIBRATION (IMPACT)

### GOOD CALIBRATION

- Uniform flip angle
- Consistent image contrast
- Accurate T<sub>1</sub>/T<sub>2</sub>/PD quantification
- Better SNR & fewer artifacts



### POOR CALIBRATION

- Non-uniform flip angle
- Signal loss / intensity variation
- Quantification errors
- Banding / shading artifacts



## 8. COMMON SOURCES OF ERROR

- Patient motion during B<sub>1</sub> map
- Wrong ROI or FOV placement
- Off-resonance (B<sub>0</sub>) inhomogeneity
- SAR limits preventing optimal pulse
- Multi-channel interference (pTx)
- Incorrect TR/TE assumptions (model-based)
- Temperature drift / system instability

### TIP

- Keep patient centered.
- Use appropriate B<sub>1</sub> method for sequence.
- Re-check after patient repositioning.
- Verify on new patients or coil changes.

## 9. PRACTICAL TARGETS & TOLERANCES

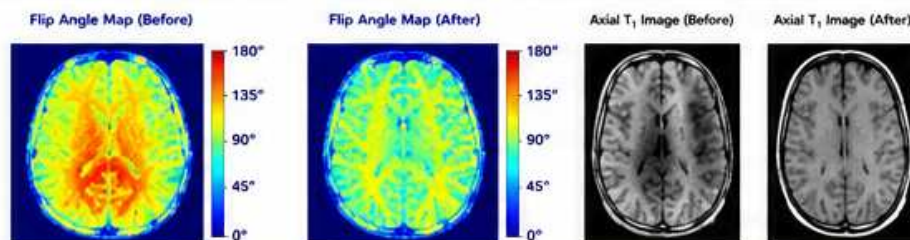
| Sequence / Purpose   | Target Flip Angle   |
|----------------------|---------------------|
| Spin Echo 90         | 90° ± 10%           |
| Spin Echo 180        | 180° ± 10%          |
| GRE (Small Flip)     | 10° – 30°           |
| Inversion Recovery   | Variable (TI based) |
| Quantitative Mapping | Per protocol        |

### Tolerance Guide

- ± 5%      Excellent
- ± 10%    Good / Acceptable
- ± 15%    Marginal
- > ± 15%   Poor (Re-calibrate)

Uniformity across ROI is as important as the average flip angle.

## 10. EXAMPLE: BEFORE & AFTER CALIBRATION (90° TARGET)



After calibration: More uniform flip angle → uniform signal → better images & quantification.

## 11. QUICK CHECK – B<sub>1</sub> CALIBRATION

- Run B<sub>1</sub> map in the imaging region.
- Check mean flip angle vs. target.
- Check uniformity (look for red/blue areas).
- Apply correction (adjust gain or pulses).
- Re-run B<sub>1</sub> map and verify.
- Document settings and results.

A good B<sub>1</sub> calibration once, saves time and improves every scan that follows.

## 12. RECOMMENDED PRACTICES

- ✓ Use correct B<sub>1</sub> mapping method for the sequence.
- ✓ Calibrate after patient positioning.
- ✓ Re-calibrate for large patient size differences.
- ✓ Monitor SAR when using high flip angles.
- ✓ Use consistent protocol & coil setup.
- ✓ Document and trend calibration results.
- ✓ Educate users: "Calibrate → Verify → Scan".

## 13. QUICK REFERENCE

| Parameter                       | Typical / Notes                   |
|---------------------------------|-----------------------------------|
| B <sub>1</sub> Field (Transmit) | RF field in $\mu T$               |
| Flip Angle ( $\alpha$ )         | Degrees (°)                       |
| 90° Pulse                       | $\alpha = 90^\circ$               |
| 180° Pulse                      | $\alpha = 180^\circ$              |
| Correction Factor (CF)          | $\alpha_{target} / \alpha_{meas}$ |
| Good Tolerance                  | ± 5–10%                           |
| Re-calibrate if                 | > ± 10–15% variation              |



### ENGINEER NOTES

- B<sub>1</sub> affects every sequence.
- Uniform excitation = uniform images.
- Calibrate often, verify every time.

### ABBREVIATIONS

B<sub>1</sub> = RF Magnetic Field (Transmit)  
 $\alpha$  = Flip Angle  
 CF = Correction Factor

ROI = Region of Interest  
 TR = Repetition Time  
 TE = Echo Time

SAR = Specific Absorption Rate  
 pTx = Parallel Transmission  
 FOV = Field of View

### SAFETY REMINDER



- High RF power can cause tissue heating.
- Monitor SAR and follow site safety procedures.
- Ensure patient comfort and communication.



### THE BOTTOM LINE

Measure B<sub>1</sub>. Match the flip angle.  
 Deliver power precisely.  
 Get uniform images. Get accurate results.